



INTERNAL

# Database Reference

Quenyx vOPS HUB — v1.0.0 · Document v2.0

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## REVISION HISTORY

| Version | Date       | Notes                                                                           |
|---------|------------|---------------------------------------------------------------------------------|
| 1.0     | 2026       | Initial v1 pack (through Sprint 19).                                            |
| 2.0     | 2026-06-29 | Aligned to v1.0.0; native QynSight check fields; Sprint 20 AI Workspace tables. |

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## 09 — Database Reference

**Audience:** Engineers, DBAs, auditors. **Source:** Derived from `backend/database/migrations` (65 migrations) and models at Sprint 19. Engine: **MySQL** via Eloquent. Regenerate this doc when migrations change.

**Conventions.** Domain entities (corpus, AI, RAG) use **UUID** identifiers. Corpus content is **immutable + provenance-tracked**. Timestamps are standard Laravel `created_at/updated_at`.

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# 1. Major table groups

| Group                 | Purpose                                                                                                  | Representative migrations                                                                                                                                                                                                      |
|-----------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Identity & tenancy    | Users, projects/workspaces, memberships, invites                                                         | <code>create_users_table</code> , <code>create_projects_table</code> , <code>create_project_members_table</code> → <code>rename_project_members_to_memberships</code> , <code>create_project_invites_table</code>              |
| Plans & subscriptions | Plans, module subscriptions, project subscriptions                                                       | <code>create_plans_table</code> , <code>create_module_subscriptions_table</code> , <code>create_project_subscriptions_table</code>                                                                                             |
| Modules & overrides   | Module catalog, per-project overrides                                                                    | <code>create_modules_table</code> , <code>create_project_module_overrides_table</code> , <code>cleanup_non_shield_modules</code>                                                                                               |
| Integrations          | Integration defs + configurations                                                                        | <code>create_integrations_table</code> , <code>create_integration_configurations_table</code> , <code>create_billing_integrations_table</code>                                                                                 |
| Audit                 | Audit log                                                                                                | <code>create_audit_logs_table</code> , <code>qcif_sprint_1_1_audit_hardening</code>                                                                                                                                            |
| QynSight (observe)    | Services, targets, hosts, alerts, metrics, port scans                                                    | <code>create_observe_services_table</code> , <code>create_observe_targets_*</code> , <code>create_observe_alert_tables</code> , <code>create_observe_metrics_history_table</code> , <code>create_host_port_scans_tables</code> |
| Agents                | Agents + enrollment tokens                                                                               | <code>create_agents_tables</code> , <code>enhance_agent_enrollment_tokens</code>                                                                                                                                               |
| Queue                 | Jobs                                                                                                     | <code>create_jobs_table</code> , <code>create_failed_jobs_table</code>                                                                                                                                                         |
| QCIF corpus           | Authorities → frameworks → releases → domains/controls/requirements, revisions, source docs, import runs | <code>create_compliance_corpus_tables</code> , <code>harden_compliance_corpus_schema</code> , <code>qcif_sprint_2_corpus_provenance_preparation</code> , <code>qcif_sprint_2a_versioning_hierarchy</code>                      |
| QCIF mapping          | Cross-framework mapping                                                                                  | <code>qcif_sprint_8_mapping_foundation</code>                                                                                                                                                                                  |
| QCIF evidence/gap/rec | Evidence, gap assessment, recommendations                                                                | <code>create_compliance_evidence_tables</code> , <code>create_compliance_gap_tables</code> , <code>create_compliance_recommendation_tables</code>                                                                              |
| AI orchestration      | Conversations/messages metadata                                                                          | <code>create_ai_orchestration_tables</code>                                                                                                                                                                                    |
| RAG vectors           | Vector index metadata                                                                                    | <code>create_rag_vector_tables</code>                                                                                                                                                                                          |

## 2. Workspace / user / project tables

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- **users** — accounts (auth via Sanctum). Profile stats added later; role/preferences columns were added then removed (see `add_role_and_preferences...` + `remove_role_and_preferences...` ).
- **projects** — the tenant/workspace entity.
- **project memberships** (renamed from `project_members` ) — user↔project with role.
- **project\_invites** — pending invitations (token-based accept).

## 3. Module / subscription tables

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- **modules** — module catalog (keys like `qynsight` , `qynshield` , ...).
- **module\_subscriptions** / **project\_subscriptions** — entitlement of modules to a project/plan.
- **project\_module\_overrides** — explicit per-project access overrides (audited).
- **plans** — subscription plans.

## 4. QynSight tables ( `observe_*` , **agents** )

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- **observe\_services** / **observe\_service\_definitions** — native service-check definitions and instances (check/retry intervals, `engine_key = 'native'` ). Exit-code semantics follow the standard 0/1/2/3 = OK/Warning/Critical/Unknown convention used by the QynSight Monitoring Engine.
- **observe\_targets** / **observe\_targets\_services** / **observe\_targets\_hosts** — monitored targets, their services, and hosts (with public IP).
- **observe\_alert tables** — alert rules, events, channels, monitoring profiles, evaluation.
- **observe\_metrics\_history** — historical metrics.
- **host\_port\_scans** — port-scan results.
- **agents** / **agent enrollment tokens** — host agents and their enrollment credentials.

## 5. QCIF tables

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Hierarchy (UUID, immutable, provenance-linked):

- **authorities** → **frameworks** → **framework\_releases** → **domains** → **controls** → **requirements**.
- **source\_documents** — official documents; imported entities carry `source_document_id` .
- **corpus\_revisions** — immutable approved snapshots; one **active** per release.
- **import\_runs** — record of each import (linked to source doc + entities).

## 6. AI orchestration tables

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- Conversation/message **metadata** tables ( `create_ai_orchestration_tables` ). **Prompt content is not stored** unless prompt logging is explicitly enabled (default off), and conversations are not persisted unless persistence is enabled (default off).

## 7. Evidence / gap / recommendation tables

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- **Evidence** — evidence records, types, statuses tied to requirements/controls.
- **Gap** — gap assessments correlating evidence to requirements (rule-derived status).
- **Recommendation** — rule-based recommendations with source rule IDs.

## 8. RAG vector metadata tables

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From `create_rag_vector_tables` :

- **rag\_vector\_indexes** — one row per indexed corpus revision (provider, model, status, provenance; UUID). **No tenant evidence by default.**
- **rag\_vector\_chunks** — chunk metadata (UUID, content hash, citation linkage). **Metadata only** — no fabricated vectors; real similarity requires a real vector backend.

## 9. Important enums

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- **Alert event status** (observe) — normalized via `fix_observe_alert_events_status_enum` .
- **Evidence status** — e.g., compliant / partially-compliant / no-evidence / expired / rejected / pending (surfaced by the executive scorecard).
- **Corpus revision status** — draft/active (immutability of active revisions).
- **Agent protocol** — migrated `nrpe` → `psap` ( `migrate_agent_protocol_nrpe_to_psap` ).

## 10. Key relationships

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